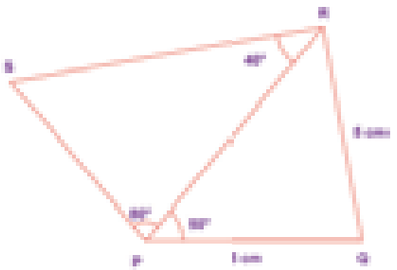
5.10.5 Construct a quadrilateral when its 2 sides and 3 angles are given

Example 5.25

Construct a quadrilateral $PQRS$ with $PQ=QR=5\ cm$, $\angle QPR = 50^\circ$, $\angle PRS = 40^\circ$ and $\angle RPS = 80^\circ$. Also find its area.

Solution:

Given:
$PQ = 5\ cm$, $QR=5\ cm$, $\angle QPR = 50^\circ$, $\angle PRS = 40^\circ$ and $\angle RPS = 80^\circ$



Rough diagram

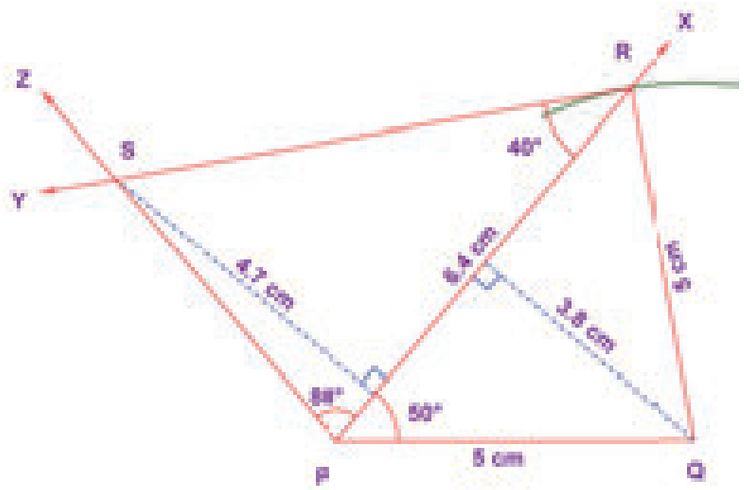


Fig. 5.44

Steps:

1. Draw a line segment $PQ = 5\ cm$.
2. At P on PQ, make $\angle QPX = 50^\circ$.
3. With Q as centre, draw an arc of radius 5 cm. Let it cut PX at R.
4. At R on PR, make $\angle PRS = 40^\circ$ and at P on PR, make $\angle RPS = 80^\circ$. Let them intersect at S.
5. $PQRS$ is the required quadrilateral.

Calculation of Area:

$$\text{Area of the quadrilateral } PQRS = \frac{1}{2} \times d \times (h_1 + h_2) \ sq.\ units$$

$$= \frac{1}{2} \times 6.4 \times (4.7 + 3.8)$$

$$= 3.2 \times 8.5 = 27.2\ cm^2$$

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